Two Particle Scattering at 1L

Calculating the Amplitudes

Elastic scattering means that the number of incoming particles = number of outgoing particles.

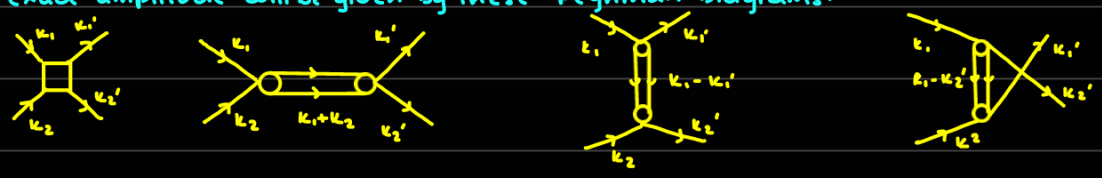
The tree-level amplitude for this process in the φ^3 theory is:

$$i\mathcal{M}_{tree} = \frac{1}{i}(ig)^2\left[\tilde{\Delta}(-s) + \tilde{\Delta}(-t) + \tilde{\Delta}(-u)\right]$$

→ Free-Field Propagator

Mandelstam Variables

The exact amplitude will be given by these Feynman Diagrams:



External lines will only contribute a factor of 1 as each external propagator contributes only the residue of the pole at $k^2=-m^2$ and this residue is 1.

We can transcribe the above Feynman Diagrams to get:

$$i\mathcal{M}_{1\text{-loop}} = \frac{1}{i}\left\{[iV_3(s)]^2\tilde{\Delta}(-s) + [iV_3(t)]^2\tilde{\Delta}(-t) + [iV_3(u)]^2\tilde{\Delta}(-u)\right\} + iV_4(s,t,u)$$

Where we have used: $\Delta(-s) = \dfrac{1}{-s+m^2-\Pi(-s)}$; $\Pi(-s) = \dfrac{1}{2}\alpha\displaystyle\int_0^1 dx\, D_2(s)\ln\left(\frac{D_2(s)}{D_0}\right) - \frac{1}{12}\alpha(-s+m^2)$

$\alpha = g/4\pi^3$

$$\frac{V_3(s)}{g} = 1 - \frac{1}{2}\alpha\int dF_3 \ln\left(\frac{D_3(s)}{m^2}\right) \quad ; \quad V_4(s,t,u) = \frac{1}{6}g^2\alpha\int dF_4\left[\frac{1}{D_4(s,t)} + \frac{1}{D_4(t,u)} + \frac{1}{D_4(u,s)}\right]$$

And we have defined: $D_2(s) = -x(1-x)s + m^2$; $D_0 = [1-x(1-x)]m^2$; $D_3(s) = -x_1x_2s + [1-(x_1+x_2)x_3]m^2$

$D_4(s,t) = -x_1x_2s - x_3x_4t + [1-(x_1+x_2)(x_3+x_4)]m^2$

We can obtain these vertex functions by quickly swapping in $k_3 = -k_1'$, $k_4 = -k_2'$ and using the definition of the mandelstam variables.

These equations are quite the handful to take in so lets try making some sense of them by taking the limit of high energy and fixed scattering angles. → Taking the limit $m^2 \to 0$

The self-energy then becomes:

$$\Pi(-s) = -\frac{1}{2}\alpha s\int_0^1 dx\, x(1-x)\left[\ln\left(\frac{-s}{m^2}\right) + \ln\left(\frac{x(1-x)}{1-x(1-x)}\right)\right] + \frac{1}{12}\alpha s = -\frac{1}{12}\alpha s\left[\ln\left(\frac{-s}{m^2}\right) + 3 - \pi\sqrt{3}\right]$$

This gives us the propagator as:

$$\Delta(-s) = \frac{1}{-s-\Pi(-s)} = \frac{-1}{s}\left(1+\frac{1}{12}\alpha\left[\ln\left(\frac{-s}{m^2}\right)+3-\pi i 3\right]\right)+O(\alpha^2)$$

We can find the appropriate branch of the log by replacing s by $s+i\epsilon$. $\because -s$ lies just below the negative real axis:

$\ln(-s) = \ln s - i\pi$

Similarly for t: $\ln(-t) = \ln|t|$, $\ln t = \ln|t| + i\pi$ $\longrightarrow$ since t is negative.

Q: How have we started calling $\Pi(-s)$ the self energy? What is the point of the fixed scattering angle limit??

What is a branch of the logarithm?

Similarly, for the three-point vertex:

$$\frac{V_2(s)}{g} = 1-\frac{1}{2}\alpha\int dF_3\left[\ln\left(\frac{-s}{m^2}\right)+\ln(x_1x_2)\right] = 1-\frac{1}{2}\alpha\left[\ln\left(\frac{-s}{m^2}\right)-3\right]$$ $\longrightarrow$ same stuff about branch.

For the four point vertex:

$$\int\frac{dF_4}{a_4(s,t)} = \frac{-g}{s+t}\left(\pi^2+\left(\ln\left|\frac{s}{t}\right|\right)^2\right) = \frac{g}{u}\left(\pi^2+\left[\ln\frac{s}{t}\right]^2\right)$$

Putting this together gives us:

$$\mathcal{T}_{1\text{-loop}} = g^2\left[F(s,t,u)+F(t,u,s)+F(u,s,t)\right]$$

where, $F(s,t,u) \equiv -\frac{1}{s}\left(1-\frac{11}{12}\alpha\left[\ln\left(\frac{-s}{m^2}\right)+c\right]-\frac{1}{2}\alpha\left[\ln\frac{t}{u}\right]^2\right)$

Loop corrections to amplitudes are typically logarithmic corrections of the kinematic variables.